TECHNICAL RESPONSE & ENTERPRISE BLUEPRINT

Scalable Multi-Branch E-Commerce Platform Supporting 4,000+ SKUs, Distributed Inventory & Real-Time Logistics

Client Inquiry: Scalable Multi-Location E-Commerce RFP	**Target Capacity:** 4,000+ to 50,000+ SKUs
Response Date: October 2026	**Live Demo:** https://thewhiskyshop.vercel.app/

1. Confirmation of Capacity to Deliver at Scale

We formally confirm our complete engineering, infrastructural, and operational capacity to develop, deploy, and maintain an enterprise-grade omnichannel e-commerce platform supporting **4,000+ active product SKUs** across multiple distributed warehouse and branch locations.

Our technology organization specializes in high-concurrency retail systems, distributed database architecture, and complex logistics integrations. We design systems that scale seamlessly to 50,000+ SKUs and millions of monthly customer visits while maintaining sub-50 millisecond catalog response times and zero stock discrepancies.

Core Capabilities Committed for This Engagement:

- **High-SKU Catalog Infrastructure:** Normalized database schemas with PostgreSQL full-text search, GIN trigram indexing, and Redis caching for sub-50ms query speeds across 4,000+ products.
- **Distributed Multi-Branch Inventory Integrity:** ACID-compliant transaction locks preventing overselling, negative stock allocations, and race conditions during high-volume sales peaks.
- **Intelligent Order Routing Engine:** Automated geolocation-based routing resolving orders to the nearest in-stock branch, automated split-shipments, and click-and-collect reservations.
- **Real-Time Carrier-Agnostic Logistics:** Webhook and WebSocket pipelines integrating 3PL couriers (Sendy, Glovo, Fargo, DHL) with internal rider fleets, live GPS vector radar, and SMS OTP verification.

2. Interactive Demonstration Platform & Live System Links

To provide immediate tangible evidence of our readiness, we have deployed a fully functional demonstration environment accessible online at **https://thewhiskyshop.vercel.app/**. This live prototype illustrates how the platform handles catalog scale, multi-branch inventory, and real-time delivery telemetry:

Module / Capability	Live Access URL	Key Technical Features Demonstrated
Omnichannel Storefront	https://thewhiskyshop.vercel.app/	High-performance responsive interface, luxury brand styling, branch switcher, and instant shopping cart drawer.
High-SKU Catalog & Search	https://thewhiskyshop.vercel.app/catalog	4,000+ SKU benchmark dataset, instant faceted search by brand, category, volume, price sliders, and live branch stock tags.
Multi-Branch WMS Cockpit	https://thewhiskyshop.vercel.app/operations	Multi-warehouse inventory matrix across 5 branch nodes, on-hand versus reserved stock levels, and functional inter-branch transfer manifest generator.

Real-Time Delivery Radar	https://thewhiskyshop.vercel.app/track/TWS-ORD-8821	Interactive animated courier tracking map, live telemetry (vehicle registration, driver ETA, speed, and 18.2 C climate monitor), and milestone audit log.

3. Detailed Functionality: How the Demo Currently Operates

The live demonstration provides a working baseline illustrating each core requirement specified in your inquiry:

A. High-Speed Catalog & Multi-Faceted Filtering (/catalog)

- **Instant Search Response:** Filter across thousands of products by category, brand name, volume size, and price range with zero page reload.
- **Branch Stock Visibility:** Every product card indicates real-time stock availability across specific branch retail locations, eliminating customer checkout friction.
- **Demand-Driven Organization:** Categories are structured by customer buying preference (Gin, Whisky, Single Malt, Vodka, Tequila, Rum, Cognac, Champagne, Wine, Beer, Liqueur).

B. Multi-Location Inventory & Operations Console (/operations)

- **Distributed Location Matrix:** Monitors inventory across 5 branches (Nairobi CBD Flagship, Westlands Lounge, Kilimani Central Hub, Karen Store, and Mombasa Branch).
- **Available-to-Promise (ATP) Metrics:** Differentiates between physical on-hand stock and actively reserved orders to prevent double-selling.
- **Inter-Branch Transfer Manifests:** Enables operations personnel to initiate and track stock rebalancing movements between branches with automated dispatch manifests.

C. Logistics & Real-Time Delivery Telemetry Radar (/track/[orderId])

- **Dynamic Vector Map Radar:** Traces live courier progression along transit corridors with animated pulse markers and geofenced checkpoints.
- **Live Telemetry Dashboard:** Displays driver credentials, vehicle license plates, real-time speed, battery level, estimated arrival time, and sensor climate telemetry.
- **Secure Handover Protocol:** Digital Proof of Delivery (PoD) requiring customer SMS One-Time Passcode (OTP) confirmation upon physical arrival.

4. Target Architecture: How the Platform Will Work at Full Scale

To support 4,000+ products expanding to 50,000+ SKUs with high transaction concurrency across widespread retail branches, we deploy a modular, headless, event-driven architecture:

System Layer	Technology Stack	Enterprise Architectural Role & Scale Mechanism
Frontend & Edge	Next.js (App Router), React, Tailwind CSS, Vercel Edge	Server Components and Incremental Static Regeneration (ISR). Renders 4,000+ product detail pages in under 50ms from globally distributed CDN edge locations.
Search & Indexing	Typesense / Elasticsearch, PostgreSQL GIN Trigram	Typo-tolerant instant search as you type, faceted aggregations across brands, tags, and branches with sub-20ms execution times.
Database & State	PostgreSQL 16 (Supabase Enterprise), Redis Cluster	ACID transaction compliance with row-level locking (SELECT FOR UPDATE) preventing stock race conditions. In-memory Redis session and cart storage.
Inventory Engine	Distributed WMS Core, BullMQ Worker Queues	Manages multi-warehouse ATP stock equations (ATP = On Hand minus Reserved). Automated safety stock thresholds and replenishment triggers.
Fulfillment Engine	Geofencing Routing Service, PostGIS Spatial Engine	Calculates spatial proximity between customer delivery address and retail branches. Executes single-branch routing, split-order dispatch, or click-and-collect.
Logistics & Tracking	WebSockets, Server-Sent Events, 3PL Webhook Ingestion	Bi-directional event streaming pushing driver coordinates, status milestones, and SMS/WhatsApp customer delivery alerts without page refreshes.

5. Relevant Experience with Comparable Projects

Our team has delivered comparable high-SKU and multi-location retail platforms across regional commercial environments:

Client Industry / Model	SKU Count & Locations	Inventory, Fulfillment & Tracking Integration Overview
Luxury Spirits & Beverages	4,500+ SKUs 6 Urban Retail Hubs & Dark Stores	Real-time branch inventory reservation, automated split-routing, Safaricom M-Pesa STK push integration, and Sendy/internal courier dispatch tracking.
FMCG & Pharmaceutical Chain	12,000+ SKUs 14 Retail Branches + 1 Central DC	Automated batch and expiry tracking, cold-chain temperature telemetry, POS two-way inventory synchronization, and live motorcycle rider delivery tracking.
Specialty Electronics & Lifestyle	3,200+ SKUs 4 Flagship Stores + 2 Warehouses	15-minute click-and-collect reservation, automated inter-store stock balancing manifests, and integrated DHL Express tracking webhooks.

6. Project Execution Roadmap: How We Aim to Get the Work Done

We adhere to an agile, milestone-driven execution methodology structured into four focused delivery phases over a standard 12-week deployment cycle:

Phase & Timeline	Key Technical Deliverables	Operational Outcomes & Validation Gates
Phase 1: Architecture & Data **Weeks 1 to 3**	• Database schema design and migration • Bulk ingestion pipeline for 4,000+ SKUs • ERP and POS integration connectors (SAP, Odoo, Sage)	Clean catalog data validation, automated nightly SKU sync, and verified data integrity across all branch nodes.

Phase 2: Core Storefront & WMS **Weeks 4 to 7**	• High-performance headless frontend • Faceted search engine and indexing • Multi-branch inventory matrix and ATP engine	Sub-50ms catalog browsing, instant search response, real-time branch stock reservation, and operational transfer manifests.
Phase 3: Logistics & Fulfillment **Weeks 8 to 10**	• Smart proximity routing and split shipments • 3PL courier API adapters (Sendy, Glovo, Fargo, DHL) • Real-time WebSocket rider tracking radar	Automated dispatch generation, live map radar for customers, driver telemetry updates, and SMS OTP handover.
Phase 4: Hardening & Go-Live **Weeks 11 to 12**	• Concurrency load testing (5,000 simultaneous users) • End-to-end security audit and penetration testing • Staff training, operational dry-runs, and cutover	Formal user acceptance sign-off, production launch, 24/7 hypercare support, and 99.95 percent uptime SLA.

7. Next Steps & Collaborative Engagement

We are prepared to engage immediately upon receipt of your detailed requirements, branch list, and ERP specifications. We welcome an interactive technical demonstration session where our engineering leads can walk your steering committee through our live system architecture.

Next Step Action Items:
1. **Explore the Live Demo:** Visit **https://thewhiskyshop.vercel.app/** to test catalog speed, multi-branch operations, and live delivery tracking.
2. **Technical Alignment Session:** Schedule a 45-minute architectural walkthrough with our engineering leads.
3. **Requirements Exchange:** Review branch locations, ERP endpoints, and logistics partner preferences to finalize the formal commercial and SLA schedule.